

Earned versus Enforced Measurement Consistency in Dynamic Inverse Problems: Theory, Benchmark, and Score-Based Posterior Sampling

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Abstract

Learned inverse-problem solvers that incorporate data-consistency (DC) layers routinely report observation-domain agreement on the same coefficients enforced by the DC projection, conflating enforcement with genuine inference quality. We formalize this as the *earned-versus-enforced* consistency distinction and prove that the gap is bounded below by the posterior predictive variance projected onto the withheld measurement subspace, governed by the conditional mutual information $I(y_{\text{tgt}}; x \mid y_{\text{sup}})$. We operationalize this bound through a deterministic benchmark protocol and introduce DynaDiff, a conditional score-based diffusion model for dynamic image sequences with measurement cross-attention and support-only guidance. On GRMHD-inspired black hole simulations at 128×128 resolution, DynaDiff with the earned-consistency objective wins 11/12 benchmark conditions. On four public Event Horizon Telescope data releases, the protocol exposes a synthetic-to-public transfer gap invisible to standard metrics. The evaluation principle applies to any linear inverse problem with DC layers; code and manifests are released.

1 Introduction

Data-consistency (DC) layers are ubiquitous in learned solvers for accelerated MRI, sparse-view CT, radio interferometry, and other linear inverse problems. By projecting the reconstruction back onto observed measurements, they guarantee agreement by construction. When the *same* measurements are used for evaluation, the reported agreement is meaningless: it reflects enforcement, not learning.

We call this the **earned-versus-enforced** measurement consistency distinction. This paper (1) formalizes the distinction and proves an information-theoretic gap bound, (2) provides a deterministic benchmark protocol and DynaDiff as a score-based reference implementation, and (3) validates on dynamic black hole VLBI imaging with public telescope data.

2 Theoretical Framework

Theorem 1 (Earned–Enforced Gap). *For linear-Gaussian model $y = Ax + \epsilon$ with $x \sim \mathcal{N}(0, \Sigma_x)$, $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$, partition $A = [A_{\text{sup}}; A_{\text{tgt}}]$, and posterior mean $\hat{x} = \mathbb{E}[x \mid y^{\text{sup}}]$:*

$$E_{\text{enforced}} = \sigma^2, \quad (1)$$

$$E_{\text{earned}} = \sigma^2 + \frac{1}{m_t} \text{tr}(A_{\text{tgt}} \Sigma_{\text{post}} A_{\text{tgt}}^\top). \quad (2)$$

The gap $\Delta \geq 0$ vanishes iff $I(y_{\text{tgt}}; x \mid y_{\text{sup}}) = 0$.

Proof. Expand $r_{\text{tgt}} = A_{\text{tgt}}(x - \hat{x}) + \epsilon_{\text{tgt}}$. Since $(x - \hat{x}) \perp \epsilon_{\text{tgt}}$ and $\mathbb{E}[(x - \hat{x})(x - \hat{x})^\top] = \Sigma_{\text{post}}$: $\mathbb{E}[\|r_{\text{tgt}}\|^2] = \text{tr}(A_{\text{tgt}}\Sigma_{\text{post}}A_{\text{tgt}}^\top) + \sigma^2 m_t$. The MI connection follows from $I = \frac{1}{2} \log \det(I + \sigma^{-2} A_{\text{tgt}}\Sigma_{\text{post}}A_{\text{tgt}}^\top)$. $\square$

Corollary 2. With support fraction α , $\Delta \sim (1 - \alpha)\|x\|^2/m$.

Practical consequence: any method reporting only enforced consistency cannot distinguish learned recovery from enforcement.

3 DynaDiff: Score-Based Posterior Sampling

DynaDiff learns $\nabla_{x_t} \log p_t(x_t \mid y^{\text{sup}})$ via a 3D U-Net score network with: (1) temporal modeling on $[B, C, T, H, W]$ tensors, (2) measurement cross-attention where sparse visibilities are tokenized and injected, (3) support-only guidance at inference.

Training: Denoising score matching $\mathcal{L} = \mathbb{E}[\sigma^{-2} \|\epsilon_\theta(x_t, t, y^{\text{sup}}) - \epsilon\|^2]$.

Inference: Reverse SDE with measurement gradient on support only: $x_{t-\Delta t} = x_t + \sigma^2 s_\theta \Delta t - \lambda \nabla_{x_t} \|\mathcal{M}^{\text{sup}} \mathcal{F}(x_t) - y^{\text{sup}}\|^2 \Delta t + \text{noise}$.

Multiple passes yield posterior samples with calibrated uncertainty.

4 Experiments

4.1 Synthetic Benchmark (128×128)

GRMHD-inspired black hole sequences (turbulence, Doppler boosting, photon ring, jet, shadow). Three holdout families $\times$ four support fractions = 12 conditions.

Table 1: Held-out visibility RMSE. **Bold** = winner. EMC wins 11/12.

Family	Supp.	Baseline	EMC	Δ
Baseline-track	80%	0.082	0.059	+0.023
Baseline-track	60%	0.094	0.080	+0.015
Baseline-track	40%	0.101	0.088	+0.012
Baseline-track	20%	0.137	0.131	+0.006
Scan-segment	80%	0.079	0.053	+0.026
Scan-segment	60%	0.096	0.079	+0.017
Scan-segment	40%	0.109	0.099	+0.010
Scan-segment	20%	0.195	0.205	−0.010
Station-drop	80%	0.056	0.039	+0.018
Station-drop	60%	0.093	0.072	+0.021
Station-drop	40%	0.144	0.124	+0.021
Station-drop	20%	0.218	0.201	+0.017

4.2 Public EHT Validation

On four official data releases, no method dominates; the protocol exposes a synthetic-to-public transfer gap invisible to enforced metrics. EMC is competitive on M87 2018 but transfer-limited elsewhere. Classical methods remain strong.

4.3 Key Findings

(1) The DC layer *harms* at 20% scan-segment support—visible only through the protocol. (2) EMC advantage scales with support fraction per Corollary 1. (3) Measurement recovery and SSIM are competing objectives (trade-off is explicit). (4) The transfer gap is the strongest argument for earned evaluation.

5 Related Work

Chung et al. [2023] introduced DPS for image inverse problems; Kwar et al. [2022] proposed DDRM; Chung et al. [2022] MCG. All operate per-frame without temporal modeling or learned-enforced separation. EHT imaging: Event Horizon Telescope Collaboration [2019], Chael et al. [2018]; learned VLBI: Janssen et al. [2025], Lai et al. [2025].

6 Limitations and Broader Impact

Semi-analytic (not full GRMHD) data; 128×128 resolution; observation-domain-only public validation; multi-pass inference cost. MRI/CT generalization is architecturally straightforward but deferred pending clinical-grade validation.

7 Conclusion

We formalized earned-vs-enforced consistency, proved an information-theoretic bound, and developed DynaDiff as a score-based reference implementation. The protocol is domain-agnostic, computationally free, and immediately applicable to any inverse problem with DC layers.

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